

On Stability of Koopman Operator-Based Output-Driven Control

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Abstract—We present conditions of stability for an output feedback control framework of (unknown) nonlinear systems using Koopman operator theory, integrating derivative estimation via high-gain observers (HGOs) and gain synthesis through linear quadratic tracking (LQT) optimal controller. We show that this modeling process can be done effectively using output data only, the whole control strategy is *output-driven*. Closed-loop stability is proven under conditions where the residual errors from the HGO estimation and the Koopman model truncation effects are bounded. The accuracy of the Koopman model depends solely on these residuals, relying instead on the quality of the data-driven model and observer performance. Numerical simulations of a nonlinear system validate the theoretical results.

I. INTRODUCTION

Koopman-based control has mainly focused on state feedback configurations, leveraging data-driven approximations of the operator to design stabilizing laws [1]. While these approaches demonstrate robustness to model uncertainties, they implicitly assume full state accessibility, which can be an impractical constraint for many physical systems where full state measurements are not possible. Recent extensions to observer design, such as [2], utilize Koopman embeddings for fault detection but do not establish closed-loop stability guarantees when integrating estimation and control. Theoretical analyses in [3] provide stability criteria for Koopman-based representations of analytic vector fields, yet their reliance on infinite-dimensional embeddings complicates practical implementation. These limitations show the need for a framework or criteria that bridges output feedback synthesis, finite-dimensional approximations, and provable stability guarantees.

Koopman-based models have outperformed neural networks in predicting pneumatic soft arm dynamics [4], enabled real-time model predictive control (MPC) for continuum manipulators [5], and accelerated shape control by a factor of 12 compared to physics-based simulations [6]. These successes stem from the operator's ability to capture non-linear phenomena through linear embeddings, even in systems with memory-dependent behaviors such as piezoelectric hysteresis [7]. Beyond robotics, the framework has provided non-conservative estimates of stability regions [8] and facilitated data-driven stability analysis in power networks [9].

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Most approaches rely on full-state measurements or assume perfect observables, as seen in MPC implementations [5], [6] and deep learning hybrids [10]. Recent advances in data-driven reinforcement learning of the Koopman space [11] and bilinear representations with finite-gain stability [12] prioritize optimality over practical sensing constraints. Similarly, model-free learning methods [13], [14] provide great techniques but often lack robustness guarantees. Through Koopman, a linear representation of the system can be found via the observer state and input matrices. With these matrices, the linear quadratic tracking (LQT) algorithm can determine the required feedback gains onto the observer states (such as the high-gain observer (HGO) output) to enable tracking. [15] introduces an integral reinforcement learning strategy to solve the LQT problem online without full dynamic models, enabling adaptive control synthesis through real-time data.

In this paper, we build on the recently introduced idea [16] of output-driven Koopman-LQT control of nonlinear systems with full-relative degree. The output-driven Koopman-LQT control architecture incorporates estimates from HGO in both the observable construction and the control feedback. Our approach is based on three key points: First, HGOs provide derivative estimates with convergence rates sufficient for closed-loop stability [2]. Second, the LQT cost structure naturally penalizes these error sources when designing feedback gains. Third, combining HGO estimates and LQT based on the Koopman model, a developed Koopman-HGO-LQT-based control is implemented. From these points, based on the perturbation analysis [17], we show the ultimate boundedness, which leads to the proof of stability in the case of vanishing perturbations.

II. PROBLEM FORMULATION

Consider a nonlinear single-input system of the form

$$\dot{x} = f(x, u), \quad y_1 = h(x), \quad (1)$$

where $x \in \mathbb{R}^n$ is the state, $u \in \mathbb{R}$ is the control input, f is assumed to be C^p smooth (continuously differentiable up to order p) nonlinear (unknown) function, where $p \geq n + 1$. $f(0, 0) = 0$ and $y_1 \in \mathbb{R}$ is the measured output specified by the (unknown) function $h(x)$. Recall that a system's relative degree, denoted by r_o , with respect to the output y_1 is the smallest integer such that the r_o -th time derivative of y_1 explicitly depends on the input u . We make the following assumption:

Assumption 1: The system (1) has a full relative degree $r_o = n$.

Many applications satisfy this assumption, such as pendulum systems and flexible joint manipulators [17].

The control objective is for the measured output y_1 to track a desired bounded reference trajectory $y_d(t)$. This objective is pursued by minimizing the infinite-horizon discounted cost function:

$$J = \frac{1}{2} \int_t^\infty e^{-\gamma(\tau-t)} [(y_1(\tau) - y_d(\tau))^2 \mathcal{Q} + u(\tau)^2 R] d\tau \quad (2)$$

where $\mathcal{Q} > 0$ and $R > 0$ are scalar weighting factors, and $0 < \gamma \leq 1$ is a discount factor where $u(\tau)$ is applied for all $\tau \geq t$. Crucially, we aim to achieve this tracking objective using only measurements of $y_1(t)$, without requiring knowledge of the original state x or the exact functions f and h . We aim to solve this problem in three steps. First, we will identify an approximate linear model using Koopman operator theory via the extended dynamic mode decomposition (EDMD) method, based on data derived from y_1 . Second, we will design an LQT state-feedback controller based on this model. Finally, we will combine the LQT controller with a HGO providing estimates of the output derivatives, leading to an output feedback control system architecture. The latter step is further motivated by the fact that HGOs satisfy the separation principle [18].

III. PRELIMINARIES

Assumption 1, essentially, implies that there exists a diffeomorphism $y = \Phi(x)$, where $y = [y_1, y_2, \dots, y_n]^T$ with $y_i = L_f^{i-1} h(x)$ for $i = 1, \dots, n$ that transforms the system into the normal (chain-of-integrator) form locally or globally, depending on the properties of f and h [17]. Here, $L_f h(x) = \frac{\partial h}{\partial x} f(x)$ denotes the Lie derivative of h along f . This form is given by:

$$\begin{aligned} \dot{y}_1 &= y_2, \\ \dot{y}_2 &= y_3, \\ &\vdots \\ \dot{y}_{n-1} &= y_n, \\ \dot{y}_n &= g(y, u), \end{aligned} \quad (3)$$

where $g(y, u)$ is a nonlinear function and $g(0, 0) = 0$. The state y consists of the output y_1 and its first $n - 1$ time derivatives along the system's vector field f .

Note: that $y \rightarrow 0$ implies $x \rightarrow 0$. $\Phi(x)$ is a diffeomorphism, therefore its inverse $\Phi^{-1}(y)$ exists and is continuous. Given the origin is an equilibrium point such that $\Phi(0) = 0$, the continuity of the inverse map ensures that as $y \rightarrow 0$, the original state $x = \Phi^{-1}(y)$ must converge to $\Phi^{-1}(0) = 0$.

The Koopman operator provides a linear representation of nonlinear dynamics (1) by lifting the state x into an infinite-dimensional space of observable functions $\psi : \mathbb{R}^n \rightarrow \mathbb{C}$ [19]. For practical computation, we approximate its action on a finite-dimensional subspace using EDMD [20].

We select a vector of q state observable functions $\Psi_x(x) = [\psi_1(x), \dots, \psi_q(x)]^T$, where the dimension q can be chosen, satisfying $q \geq n$. For systems transformable into the normal

form (3), a common strategy is to align these observables with the normal form state y , aiming for $\Psi_x(x) \approx y = \Phi(x)$. For clarity of presentation in this paper, we focus on the case where $q = n$, constructing Ψ_x based on the measured output y_1 and estimates of its first $n - 1$ derivatives via HGO, such that $\Psi_x \approx [y_1, y_2, \dots, y_n]^T$. This choice has demonstrated practical utility in generating useful models [16]. It is worth mentioning that choosing $q > n$, by including higher-order derivatives or nonlinear combinations of the base states (y_i) as additional observables, can lead to a more accurate Koopman linear model [16], [20]. This improvement comes from richer information that can potentially provide a better approximation to an invariant subspace of the Koopman operator, thereby reducing the truncation error component (e_{trunc}) of the model residual, see Section IV for further explanation. While specific error bounds often depend on the chosen basis and system properties, increasing the dimension q can improve model accuracy. The theoretical framework and stability results developed in this paper for $q = n$ can be extended to cases where $q > n$, affecting the dimensions of the Koopman model.

Given M data snapshots $\{(\Psi_{x_k}, u_k, \Psi_{x_{k+1}})\}_{k=0}^{M-1}$, where Ψ_{x_k} is the state observable vector at time t_k and $\Psi_{x_{k+1}}$ is the vector at time $t_k + \Delta t$ under input u_k , we construct the augmented observable vector including input:

$$\Psi_k = \begin{bmatrix} \Psi_{x_k} \\ u_k \end{bmatrix} \in \mathbb{R}^{q+1} \quad (4)$$

EDMD computes an approximate Koopman matrix $\hat{K} \in \mathbb{R}^{(q+1) \times (q+1)}$ via least-squares regression that minimizes $\|\Psi_{x_{k+1}} - \hat{K}\Psi_k\|^2$ over the dataset (see [16], [20]). Partitioning $\hat{K}$ yields a discrete-time linear model for the state observables Ψ_x :

$$\Psi_{x_{k+1}} \approx A_d \Psi_{x_k} + B_d u_k, \quad (5)$$

where $A_d \in \mathbb{R}^{q \times q}$ and $B_d \in \mathbb{R}^{q \times 1}$ are extracted from $\hat{K}$. This discrete-time model (A_d, B_d) is converted to a continuous-time representation $(\tilde{A}, \tilde{B})$ using a suitable transformation, e.g., the Tustin method with sampling period $T = \Delta t$ [21]. This yields the approximate continuous-time Koopman linear model describing the evolution of the chosen state observables Ψ_x :

$$\dot{\Psi}_x(t) \approx \tilde{A} \Psi_x(t) + \tilde{B} u(t). \quad (6)$$

This linear model forms the basis for our subsequent LQT controller design.

IV. ERROR ANALYSIS AND STABILITY FRAMEWORK

The identified Koopman model (6) provides an approximation to the true dynamics of the observables of the chosen state Ψ_x . Let $\dot{\Psi}_x(t)$ denote the exact time derivative of $\Psi_x(x(t))$ evolving along the trajectories of the original nonlinear system (1). The discrepancy between the true evolution and the prediction of the linear model is captured by the residual error of the Koopman model $r(t)$. Therefore,

the exact dynamics of the Koopman state observables Ψ_x are governed by:

$$\dot{\Psi}_x(t) = \tilde{A}\Psi_x(t) + \tilde{B}u(t) + r(t). \quad (7)$$

This residual $r(t)$ represents the inherent error in the finite-dimensional linear approximation obtained via EDMD. Following established analyses [20], [22], [23], this error arises primarily from:

- 1) *Discretization Error (e_{disc})*: Introduced by the discrete to continuous time conversion. For Tustin, this error is typically bounded proportionally to the square of the sampling period T , giving $\|e_{disc}\| \leq \frac{T^2}{12} \left\| \frac{d^3 \Psi_x}{dt^3} \right\| + \mathcal{O}(T^3)$ [24].
- 2) *Observable Truncation Error (e_{trunc})*: Results of projecting infinite-dimensional Koopman dynamics onto the finite subspace spanned by Ψ_x . This reflects the inability of the basis to form a perfectly invariant subspace and includes numerical errors from the regression used to compute $\hat{K}$.

For stability analysis, we will consolidate these effects and assume that the combined residual r can be bounded, relative to the Koopman state magnitude, within a region of operation.

$$\|r(t)\| \leq \delta_\Psi \|\Psi_x(t)\| + \delta_0, \quad (8)$$

for some nonnegative constants δ_Ψ and δ_0 , representing state-dependent and constant error components, respectively. For example, δ_Ψ could capture scaling truncation errors with the state, while δ_0 might include discretization errors (assuming bounded higher derivatives) and regression noise. Bounds often depend on the "energy" in the neglected observables or higher-order terms, potentially related to the norm of the state or observables themselves [22]. For example, if higher-order terms $\psi_i(x)$ are neglected ($i > q$), the error can be assumed bounded as $\|e_{trunc}\| \leq \sum_{i=q+1}^N \mathcal{L}_i \|\psi_i(x)\|$, where $\mathcal{L}_i$ is a constant associated with the i^{th} element.

We are ultimately interested in the asymptotic behavior of the system (1), or equivalently (3). Towards this goal, consider (3) and (7) and define the state approximation error:

$$e_y(t) := y(t) - \Psi_x(t). \quad (9)$$

Analyzing the stability of e_y or y directly remains challenging due to the presence of the unknown nonlinearity $g(y, u)$ and the coupled dependence on Ψ_x . Our strategy is therefore to design the controller based on the linear Koopman model for Ψ_x and analyze the stability of Ψ_x itself, treating r as a bounded disturbance. The stability of the equilibrium point of (1) then relies on the premise that if the stability of the equilibrium point of Ψ_x is stabilized and the approximation error e_y is small (meaning Ψ_x accurately represents y), then y will also exhibit stable behavior. This small e_y is expected since choosing observables Ψ_x to include the state and its time derivatives (targeting $y = [y_1, \dot{y}_1, \dots, y_1^{(n-1)}]^T$) makes the Koopman linear model a finite-order Taylor series approximation of the system's evolution. As demonstrated in [25], for derivative-based observables, this truncation

error can be explicitly bounded. Their analysis, leveraging the Lagrange remainder formula for Taylor series, showed that the prediction error e_k for a state after k time steps (prediction horizon $T = k\Delta t$), using n derivatives of the output as observables, is bounded by:

$$|e_k| \leq \frac{T^{n+1}}{(n+1)!} \left| y_{1_{\max}}^{(n+1)} \right|, \quad (10)$$

where $\left| y_{1_{\max}}^{(n+1)} \right|$ is the maximum magnitude of the $(n+1)$ -th derivative of the output state over the time interval. It is important to note that the error e_k is derived from data collected during the training stage, where the input u_k is chosen as an arbitrary stabilizing signal. This step allows for rich data collection, which is crucial for accurate model training.

This bound highlights that if the higher order derivatives of the system dynamics (specifically of $g(y, u)$ in normal form coordinates) are bounded, the truncation error component of the residual $r(t)$ is also bounded, and can be made small by including a sufficient number of derivatives (increasing n) or operating over shorter time horizons. If the underlying nonlinear system (1) is smooth such that its higher-order derivatives are bounded in the operation region, the approximation error e_y can be kept small by the choice of observables. Therefore, stabilizing the Koopman state Ψ_x under the condition of a sufficiently small residual r ensures the boundedness or stability of the original system state y .

A. LQT Koopman-based Feedback Control

Assuming the Koopman state observables Ψ_x are available (ideal state-feedback scenario), we design an LQT controller aimed at making the first component $\Psi_{x,1}$ (approximating y_1) track the desired reference y_d . The control input is synthesized as an affine function of the Koopman state observables and the reference. The LQT produces the following feedback gain:

$$u = K_1 \begin{bmatrix} \Psi_x \\ y_d \end{bmatrix}, \quad (11)$$

or written as:

$$u(t) = K_{\Psi_x} \Psi_x(t) + K_{y_d} y_d(t), \quad (12)$$

where the feedback gain $K_{\Psi_x} \in \mathbb{R}^{1 \times q}$ and desired output gain $K_{y_d} \in \mathbb{R}$ are computed by solving the LQT problem associated with the linear Koopman model $(\tilde{A}, \tilde{B})$ and the cost function (2). Substituting this control law into the exact dynamics of Ψ_x (7):

$$\dot{\Psi}_x = \tilde{A}\Psi_x + \tilde{B}(K_{\Psi_x} \Psi_x + K_{y_d} y_d) + r(t). \quad (13)$$

Defining the closed-loop matrix $\bar{A} := \tilde{A} + \tilde{B}K_{\Psi_x}$ and the tracking input matrix $\bar{B} := \tilde{B}K_{y_d}$, the closed-loop dynamics become:

$$\dot{\Psi}_x = \bar{A}\Psi_x + \bar{B}y_d + r(t). \quad (14)$$

To analyze stability, we assume the LQT design yields a Hurwitz matrix $\bar{A}$. We employ the quadratic Lyapunov function $V(\Psi_x) = \Psi_x^T P \Psi_x$, where $P = P^T > 0$ is the

solution to the Lyapunov equation $\bar{A}^T P + P \bar{A} = -Q$ for a chosen definite matrix $Q = Q^T > 0$. The time derivative $\dot{V}$ along the trajectories of (14) satisfies:

$$\begin{aligned}\dot{V}(\Psi_x) &= \Psi_x^T (\bar{A}^T P + P \bar{A}) \Psi_x + 2\Psi_x^T P (\bar{B}y_d + r) \\ &= -\Psi_x^T Q \Psi_x + 2\Psi_x^T P \bar{B}y_d + 2\Psi_x^T P r \\ &\leq -\lambda_{\min}(Q) \|\Psi_x\|^2 + 2\|\Psi_x\| \|P \bar{B}\| \|y_d\| \\ &\quad + 2\|\Psi_x\| \|P\| \|r\|. \end{aligned} \quad (15)$$

Using the bound $\|r(t)\| \leq \delta_\Psi \|\Psi_x(t)\| + \delta_0$ from (8):

$$\begin{aligned}\dot{V}(\Psi_x) &\leq -(\lambda_{\min}(Q) - 2\|P\|\delta_\Psi) \|\Psi_x\|^2 \\ &\quad + (2\|P \bar{B}\| \|y_d\| + 2\|P\|\delta_0) \|\Psi_x\|. \end{aligned} \quad (16)$$

This inequality allows us to establish conditions for the stability or boundedness of the Koopman state observable vector Ψ_x .

Theorem 1 (State-Driven LQT Koopman): Consider the closed-loop system for the Koopman state observables Ψ_x given by (14), using the control law (12). If the state-dependent part of the residual error is sufficiently small based on (16) such that:

$$\delta_\Psi < \frac{\lambda_{\min}(Q)}{2\|P\|}, \quad (17)$$

Given the desired reference trajectory $y_d(t)$ is bounded, $\|y_d(t)\| \leq Y_d$. The Koopman state observable vector $\Psi_x(t)$ is ultimately bounded, and its trajectories converge to the set,

$$\{\Psi_x \mid \|\Psi_x\| \leq \frac{2\|P \bar{B}\| Y_d + 2\|P\|\delta_0}{\lambda_{\min}(Q) - 2\|P\|\delta_\Psi}\}. \quad (18)$$

In the case of regulation ($y_d = 0$) and assuming the residual error vanishes at the origin ($\delta_0 = 0$), condition (17) guarantees global exponential stability of the origin $\Psi_x = 0$.

V. OUTPUT-FEEDBACK CONTROL USING HIGH-GAIN OBSERVER

When the derivatives of the output are unmeasurable, a live estimate can be derived using the HGO. However, the HGO introduces estimation errors, which, although small, must be accounted for in the control design. These errors are typically on the order of the HGO parameter ε , which can be selected to achieve the desired estimation accuracy.

In practice, only the output y_1 is available. To implement the controller (12), we require estimates of the Koopman state observables Ψ_x . Since Ψ_x is chosen to approximate the normal form state y , we can use a HGO designed for the normal form structure (3) to estimate y , and then use this estimate $\hat{y}$ as our estimate for Ψ_x . The HGO equations are [18], [26]:

$$\begin{aligned}\dot{\hat{y}}_1 &= \hat{y}_2 + \frac{\alpha_1}{\varepsilon} (y_1 - \hat{y}_1), \\ \dot{\hat{y}}_2 &= \hat{y}_3 + \frac{\alpha_2}{\varepsilon^2} (y_1 - \hat{y}_1), \\ &\vdots \\ \dot{\hat{y}}_{n-1} &= \hat{y}_n + \frac{\alpha_{n-1}}{\varepsilon^{n-1}} (y_1 - \hat{y}_1), \\ \dot{\hat{y}}_n &= \frac{\alpha_n}{\varepsilon^n} (y_1 - \hat{y}_1), \end{aligned} \quad (19)$$

where $\hat{y} = [\hat{y}_1, \dots, \hat{y}_n]^T$ is the estimate of the normal form state y . The constants $\alpha_1, \dots, \alpha_n$ are chosen such that the polynomial $s^n + \alpha_1 s^{n-1} + \dots + \alpha_n = 0$ has roots with negative real parts. The parameter $0 < \varepsilon \ll 1$ determines the observer's speed and sensitivity. The HGO estimation error $\eta(t) := y(t) - \hat{y}(t)$ converges rapidly to a small value bounded by the observer parameter ε , provided the system dynamics are well-behaved:

$$\|\eta(t)\| \leq \delta_\eta \varepsilon, \quad \text{for } t \geq T_{\text{transient}}, \quad (20)$$

where $\delta_\eta > 0$ is a constant depending on bounds on the derivatives of $g(y, u)$ [18].

We use the HGO estimate $\hat{y}$ directly as the estimate for the Koopman state observables required by the controller, i.e., $\hat{\Psi}_x(t) := \hat{y}(t)$ (assuming we chose $q = n$ and $\Psi_x \approx y$). The output-feedback LQT control law is thus:

$$u(t) = K_{\Psi_x} \hat{\Psi}_x(t) + K_{y_d} y_d(t) = K_{\Psi_x} \hat{y}(t) + K_{y_d} y_d(t). \quad (21)$$

To analyze stability, we consider the dynamics of the true Koopman state Ψ_x under this output-feedback law. Substituting (21) into the exact dynamics (7):

$$\dot{\Psi}_x = \tilde{A} \Psi_x + \tilde{B} (K_{\Psi_x} \hat{y} + K_{y_d} y_d) + r(t). \quad (22)$$

We express the HGO estimate $\hat{y}$ in terms of Ψ_x , the primary approximation error $e_y = y - \Psi_x$, and the HGO error $\eta = y - \hat{y}$: $\hat{y} = y - \eta = (\Psi_x + e_y) - \eta$. Substituting this gives:

$$\begin{aligned}\dot{\Psi}_x &= \tilde{A} \Psi_x + \tilde{B} (K_{\Psi_x} (\Psi_x + e_y - \eta) + K_{y_d} y_d) + r(t) \\ &= (\tilde{A} + \tilde{B} K_{\Psi_x}) \Psi_x + \tilde{B} K_{\Psi_x} (e_y - \eta) + \tilde{B} K_{y_d} y_d + r(t) \\ &= \bar{A} \Psi_x + \tilde{B} K_{\Psi_x} (e_y - \eta) + \bar{B} y_d + r(t). \end{aligned} \quad (23)$$

The evolution of the Koopman state Ψ_x under output feedback is now perturbed by the Koopman residual r , the HGO estimation error η , and the primary state approximation error e_y .

A. Stability Analysis

We analyze the stability of the true state y governed by (23) using the same Lyapunov function $V(y) = y^T P y$, where $\bar{A}^T P + P \bar{A} = -Q$. Starting with $y(t) = \Psi_x(t) + e_y(t)$, leads to $\dot{y}(t) = \dot{\Psi}_x(t) + \dot{e}_y(t)$. Substituting the expression for $\dot{\Psi}_x$ from (23) gives:

$$\begin{aligned}\dot{y} &= \left(\bar{A} \Psi_x + \tilde{B} K_{\Psi_x} (e_y - \eta) + \bar{B} y_d + r(t) \right) + \dot{e}_y \\ &= \bar{A} y - \tilde{B} K_{\Psi_x} \eta + \bar{B} y_d + \underbrace{\left[r(t) - (\bar{A} - \tilde{B} K_{\Psi_x}) e_y + \dot{e}_y \right]}_{\text{Lumped Residual } (\mathbf{r})}. \end{aligned} \quad (24)$$

The term in the brackets represents a lumped residual that combines the original Koopman model error $r(t)$ with the effects of the primary approximation error e_y and its dynamics. As shown by the bound in (10), both e_y and its derivatives $\dot{e}_y$ are bounded. Therefore, we can treat this entire lumped residual as a single bounded residual, $\mathbf{r}$. The time derivative

of the Lyapunov function $V(y)$ along the trajectories of (24) is:

$$\dot{V}(y) = y^T(\bar{A}^T P + P \bar{A})y + 2y^T P(-\tilde{B}K_y \eta + \bar{B}y_d + \mathbf{r}) \quad (25)$$

$$= -y^T Q y - 2y^T P \tilde{B} K_y \eta + 2y^T P \bar{B} y_d + 2y^T P \mathbf{r} \quad (26)$$

$$\leq -\lambda_{\min}(Q)\|y\|^2 + 2\|y\|\|P \tilde{B} K_y\|\|\eta\| + 2\|y\|\|P \bar{B}\|\|y_d\| + 2\|y\|\|P\|\|\mathbf{r}\|. \quad (27)$$

Using the bounds $\|\eta\| \leq \delta_\eta \varepsilon$ from (20) and $\|\mathbf{r}\| \leq \delta_\Psi \|y\| + \delta_0$ from (8):

$$\begin{aligned} \dot{V}(y) &\leq -(\lambda_{\min}(Q) - 2\|P\|\delta_\Psi)\|y\|^2 \\ &\quad + (2\|P \tilde{B} K_y\|\delta_\eta \varepsilon + 2\|P \bar{B}\|\|y_d\| + 2\|P\|\delta_0)\|y\|. \end{aligned} \quad (28)$$

This inequality provides the basis for establishing ultimate boundedness of the Koopman state Ψ_x under output feedback, explicitly showing the influence of the HGO error ($\propto \varepsilon$) and the Koopman residual (δ_Ψ, δ_0).

Theorem 2 (Output-Driven LQT Koopman): Consider the closed-loop system using the output-feedback LQT law (21) with HGO estimates derived from (19). Assume the Koopman model ($\bar{A}, \bar{B}$) was identified based on observables approximating the true states of y . Based on (28) if the state-dependent component of the Koopman residual δ_Ψ is sufficiently small, such that

$$\delta_\Psi < \frac{\lambda_{\min}(Q)}{2\|P\|}, \quad (29)$$

then the Koopman state observable vector $\Psi_x(t)$, is ultimately bounded for sufficiently small ε . Specifically, its trajectories converge to the set

$$\{\Psi_x \mid \|\Psi_x\| \leq \frac{2\|P \tilde{B} K_{\Psi_x}\|\delta_\eta \varepsilon + 2\|P \bar{B}\|Y_d + 2\|P\|\delta_0}{\lambda_{\min}(Q) - 2\|P\|\delta_\Psi}\}. \quad (30)$$

Remark 1: The size of this ultimate bound is influenced by the Koopman model's accuracy (reflected in δ_Ψ, δ_0) and the HGO performance (parameter ε). For the regulation problem ($y_d = 0, \delta_0 = 0$), the ultimate bound approaches zero as the HGO parameter $\varepsilon \rightarrow 0$, provided the stability condition on δ_Ψ holds.

This theorem provides the main stability result for the output-feedback scheme, proving ultimate boundedness for the true system state y under quantifiable conditions related to modeling and estimation errors. The result highlights the connection between modeling errors and observer precision in determining closed-loop performance.

VI. SIMULATION EXAMPLE

To validate the proposed output-driven Koopman-LQT control framework incorporating HGO estimation, we consider a two-dimensional nonlinear dynamical system described in [18]:

$$\dot{x}_1 = -x_1 + x_2 \quad (31a)$$

$$\dot{x}_2 = -x_1^3 - x_2 + u \quad (31b)$$

where $x = [x_1, x_2]^T$ is the state vector and u is the control input. The output available for measurement is $y_1 = x_1$. Initial conditions are $x_0 = [0, 0]^T$. We started data collection by using a chirp signal to cover a broad frequency range. The input u swept from 0.1 Hz to 1 Hz over a 20-second interval, as shown in Fig. 1.

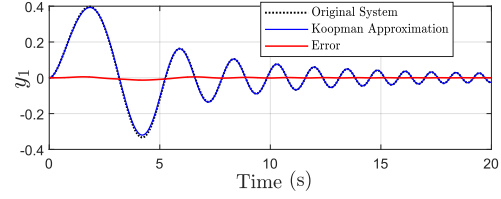


Fig. 1: Output of Koopman-based model based on data and HGO derivatives of the data. A chirp input u swept from 0.1 Hz to 1 Hz over a 20-second interval, to gather rich data.

We then apply the Koopman algorithm to the system's output y_1 , its estimated derivative $\hat{y}_1$ from HGO, and the input u to obtain the Koopman matrices, which are used to implement LQT control for the motion system.

The observer is configured to estimate up to the third derivative, enhancing the accuracy of the first derivative estimates for the training data and later for the controller. The chosen pole locations are $[-10, -15, -20, -25]$, and solving the Hurwitz polynomial yields coefficients $\alpha = [70, 1775, 19250, 75000]$ with a tuning parameter set to $\epsilon = 0.005$. We used the first $q = n = 2$ HGO states $\hat{y} = [\hat{y}_1, \hat{y}_2]^T$ as observables. EDMD followed by Tustin transformation yielded:

$$\bar{A} = \begin{bmatrix} 0.0002 & 1.0002 \\ -1.083 & -2.0613 \end{bmatrix}, \quad \bar{B} = \begin{bmatrix} -0.00013 \\ 1.008 \end{bmatrix}. \quad (32)$$

The chosen LQT parameters are $R = 0.1$, $Q = 1 \times 10^6$, and $\gamma = 0.1$, resulting in the feedback gains

$$K_1 = [-3157.2 \quad -77.487 \quad 3158.3]. \quad (33)$$

With the controller in place, we tested a sinusoidal tracking reference (y_d) with a 1-unit amplitude and 0.5 Hz frequency (see Fig. 2). The system achieved good performance, with a sinusoidal error of approximately 0.079 units or an error of 7.9%. The response matches the desired, just delayed by approximately 25 ms, resulting in such a sinusoidal error. Next, a step response is tracked where the desired output (y_d) is set to a value of 1 at $t = 2$ s. A saturation on the input of 500 is also applied as shown in Fig. 3.

VII. CONCLUSION

This paper introduces a Koopman-based output-driven feedback control framework for nonlinear systems, using derivative estimation via HGOs as the observable functions in a Koopman model and gain synthesis through LQT. By constructing observable functions from HGO-derived state estimates, the method lifts the nonlinear dynamics into a higher-dimensional linear space, enabling the application of LQT to design stabilizing feedback gains. Closed-loop stability is established under conditions where residual errors

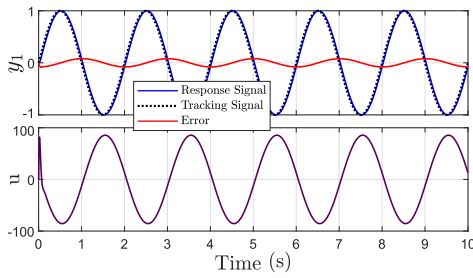


Fig. 2: Tracking of a sinusoidal objective $y_d = \sin(\pi t)$. The output-driven control is shown to track the signal within a margin of 7.9%, due to response delay.

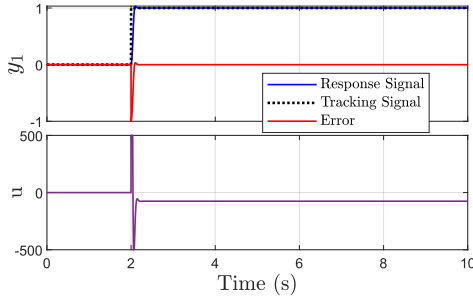


Fig. 3: Tracking of a step response $y_d = 1$, $t \geq 2$ s. The output-driven control is shown to stabilize at the desired.

from the HGO's derivative estimation and the truncation effects of EDMD remain bounded. A critical insight is that the accuracy of the Koopman model is dependent on these residuals. This work bridges data-driven modeling with classical control theory, offering a systematic approach to output feedback design for nonlinear systems without full-state measurements. To test this, a two-state nonlinear system featuring cubic nonlinearities that does not conform to the standard chain-of-integrator configuration is modeled and controlled to track a sinusoidal trajectory and a step response. The results show error reduction and stabilization. Future work includes exploring systems where the relative degree is less than the system dimension ($r < n$), alternative controller designs, and optimizing observable selection for improved performance and robustness to different trajectories.

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